

Chapter 1

The $\mathcal{N} = 2$ superconformal algebra

The $\mathcal{N} = 2$ superconformal algebra is the definitive witness in the standard landscape that chiral Koszulness and Chevalley–Eilenberg Koszulness are genuinely distinct properties: the CE complex has $H_{\text{CE}}^2 \neq 0$ at conformal weight 3, yet the chiral bar complex has $H_{\text{bar}}^2 = 0$. The sub-leading OPE modes in the chiral bar differential, invisible to the CE complex, provide coboundary contributions that kill the CE cocycles.

The Kazama–Suzuki coset decomposition $\text{SCA}_c \simeq \text{Com}(\text{U}(1), \widehat{\mathfrak{sl}}_{2,k} \otimes \psi\bar{\psi})$ determines the modular characteristic $\kappa = (k+4)/4$; the Feigin–Frenkel involution $k \mapsto -k - 4$ identifies the Koszul dual at central charge $c' = 6 - c$; the complementarity sum $\kappa(c) + \kappa(6-c) = 1$ is constant. The shadow obstruction tower has multi-channel structure: the T -channel is class M (infinite depth, as for Virasoro), the J -channel is class G (Gaussian, terminating at arity 2), and the overall shadow class is M, governed by the deepest channel. All five main theorems are verified.

Table 1.1: Five-theorem verification for the $\mathcal{N} = 2$ superconformal algebra at generic level $k \neq -2$.

Theorem	Statement for SCA_c	Status	Reference
A (adjunction)	$\text{SCA}_c^! \simeq \text{SCA}_{6-c}$ (Kazama–Suzuki involution $k \mapsto -k - 4$)	Proved	Thm ??
B (inversion)	$\Omega(\overline{B}(\text{SCA}_c)) \xrightarrow{\sim} \text{SCA}_c$	Proved	Thm ??
C (complementarity)	$\kappa(c) + \kappa(6-c) = 1$	Proved	Prop ??
D (modular char.)	$\kappa(\text{SCA}_c) = \frac{6-c}{2(3-c)} = \frac{k+4}{4}$	Proved	Prop ??
H (Hochschild)	$\text{ChirHoch}^*(\text{SCA}_c)$ polynomial	Proved	Thm ??

The modular Koszul triple (Definition ??) of the $\mathcal{N} = 2$ SCA at generic level $k \neq -2$ is

$$\mathfrak{T}_{\text{SCA}} = (\text{SCA}_c, \text{SCA}_{6-c}, r_{\text{SCA}}(z)), \quad (1.1)$$

Table 1.2: Shadow archetype data for the $\mathcal{N} = 2$ superconformal algebra.

Invariant	Value
Class	M (mixed: T -channel ∞ , J -channel G)
Shadow depth $r_{\max}$	∞
$\kappa(\text{SCA}_c)$	$(6-c)/(2(3-c)) = (k+4)/4$
Central charge	$c = 3k/(k+2)$; $c' = 6 - c$
Complementarity	$\kappa + \kappa' = 1$ (W-type)
Self-dual point	$c = 3$ ($k \rightarrow \infty$, free-field limit)
Critical level	$k = -2$: Sugawara undefined
r -matrix $r(z)$	Multi-channel: T -channel has poles at z^{-3}, z^{-1} ; J -channel has pole at z^{-1}
CE vs chiral bar	$H_{\text{CE}}^2 \neq 0$ at wt 3; $H_{\text{bar}}^2 = 0$ (chiral Koszul)

where the r -matrix has two bosonic channels: a triple pole $(c/2)/z^3 + 2T/z$ from the Virasoro subalgebra and a simple pole $(c/3)/z$ from the $U(1)$ current J .

The $\mathcal{N} = 2$ SCA occupies a distinguished position in the shadow depth classification. The Virasoro subalgebra places it in class M ($r_{\max} = \infty$), while the $U(1)$ current gives a class G channel ($r_{\max} = 2$) and the fermionic G^+G^- pairing gives a class L channel ($r_{\max} = 3$). The overall class is M, governed by the deepest channel. Extended supersymmetry modifies the modular characteristic relative to the Virasoro value: the coset decomposition (Proposition ??) yields $\kappa = (k+4)/4 = (6-c)/(2(3-c))$, which diverges at the free-field limit $c = 3$ and vanishes at $c = 6$ (corresponding to $k = -4$).

1.1 OPE data and generators

The $\mathcal{N} = 2$ superconformal algebra has four generators:

$$T \text{ (wt 2, bosonic), } J \text{ (wt 1, bosonic), } G^+ \text{ (wt } \frac{3}{2}, \text{ fermionic), } G^- \text{ (wt } \frac{3}{2}, \text{ fermionic).} \quad (1.2)$$

The stress tensor T generates a Virasoro subalgebra at central charge c ; the $U(1)$ current J has charge assignments $\text{ch}(G^+) = +1$, $\text{ch}(G^-) = -1$, $\text{ch}(T) = \text{ch}(J) = 0$; and the supercurrents $G^\pm$ are fermionic primaries of conformal weight $\frac{3}{2}$.

The OPE algebra is (OPE-mode convention: $a_{(n)}b$ denotes the coefficient of $(z-w)^{-(n+1)}$):

$$T_{(3)}T = \frac{c}{2}, \quad T_{(1)}T = 2T, \quad T_{(0)}T = \partial T, \quad (1.3)$$

$$T_{(1)}J = J, \quad T_{(0)}J = \partial J, \quad (1.4)$$

$$T_{(1)}G^\pm = \frac{3}{2}G^\pm, \quad T_{(0)}G^\pm = \partial G^\pm, \quad (1.5)$$

$$J_{(1)}J = \frac{c}{3}, \quad (1.6)$$

$$J_{(0)}G^+ = G^+, \quad J_{(0)}G^- = -G^-, \quad (1.7)$$

$$G_{(2)}^+G^- = \frac{c}{3}, \quad G_{(1)}^+G^- = J, \quad G_{(0)}^+G^- = T + \frac{1}{2}\partial J, \quad (1.8)$$

$$G_{(n)}^\pm G^\pm = 0 \quad \text{for all } n. \quad (1.9)$$

The fermionic self-OPEs vanish identically. By vertex superalgebra skew-symmetry, the reversed zero-mode product is $G_{(0)}^- G^+ = T - \frac{1}{2} \partial J$ (the sign on ∂J flips relative to $G_{(0)}^+ G^- = T + \frac{1}{2} \partial J$); the higher reversed modes $G_{(n)}^- G^+$ for $n \geq 1$ receive derivative corrections from the skew-symmetry formula.

Remark 1.1 (Kazama–Suzuki coset realization). The $\mathcal{N} = 2$ SCA at central charge $c = 3k/(k+2)$ is realized as the Kazama–Suzuki coset

$$\text{SCA}_c \simeq \text{Com}(\text{U}(1), \widehat{\mathfrak{sl}}_{2,k} \otimes \psi\bar{\psi}), \quad (1.10)$$

where $\psi\bar{\psi}$ denotes a complex free fermion pair and $\text{U}(1)$ is the diagonal current $J_{\text{diag}} = J_{\text{KM}}^3 + : \psi\bar{\psi} :$. The level k ranges over $\mathbb{C} \setminus \{-2\}$; at $k = -2$ the Sugawara construction is undefined. The free-field limit $k \rightarrow \infty$ gives $c \rightarrow 3$.

1.2 Modular characteristic

Proposition 1.2 (Modular characteristic of the $\mathcal{N} = 2$ SCA;). *At central charge $c = 3k/(k+2)$ with $k \neq -2$:*

$$\kappa(\text{SCA}_c) = \frac{k+4}{4} = \frac{6-c}{2(3-c)}. \quad (1.11)$$

Proof. The Kazama–Suzuki decomposition (??) gives κ as the coset difference of modular characteristics:

$$\begin{aligned} \kappa(\text{SCA}_c) &= \kappa(\widehat{\mathfrak{sl}}_{2,k}) + \kappa(\psi\bar{\psi}) - \kappa(\text{U}(1)_{\text{diag}}) \\ &= \frac{3(k+2)}{4} + \frac{1}{2} - \left(\frac{k}{2} + 1\right) = \frac{3k+6+2-2k-4}{4} = \frac{k+4}{4}. \end{aligned}$$

The coset constituents: $\kappa(\widehat{\mathfrak{sl}}_{2,k}) = \dim(\mathfrak{sl}_2)(k+h^\vee)/(2h^\vee) = 3(k+2)/4$; $\kappa(\psi\bar{\psi}) = 1/2$ (complex free fermion at $c = 1$); $\kappa(\text{U}(1)_{\text{diag}}) = k/2 + 1$, since the diagonal current $J_{\text{diag}} = J_{\text{KM}}^3 + : \psi\bar{\psi} :$ has level $\ell = k/2 + 1$ (the Cartan generator J^3 contributes $k/2$ and the fermion bilinear contributes 1). Substituting $k = 2c/(3-c)$ yields $(6-c)/(2(3-c))$. $\square$

1.3 Koszul duality and complementarity

Proposition 1.3 (Complementarity for the $\mathcal{N} = 2$ SCA;). *The Feigin–Frenkel involution $k \mapsto -k - 2h^\vee = -k - 4$ sends $c = 3k/(k+2)$ to $c' = 6 - c$. The modular characteristics satisfy*

$$\kappa(c) + \kappa(6-c) = 1. \quad (1.12)$$

The self-dual point is $c = 3$ ($k \rightarrow \infty$).

Proof. Under $k \mapsto -k - 4$: $c' = 3(-k-4)/(-k-2) = 3(k+4)/(k+2)$. Then $c + c' = 3k/(k+2) + 3(k+4)/(k+2) = 6$. For the complementarity sum:

$$\kappa(c) + \kappa(6-c) = \frac{6-c}{2(3-c)} + \frac{6-(6-c)}{2(3-(6-c))} = \frac{6-c}{2(3-c)} + \frac{c}{2(c-3)} = \frac{6-c}{2(3-c)} - \frac{c}{2(3-c)} = \frac{6-2c}{2(3-c)} = 1.$$

This is W-type complementarity: the sum is 1, not 0 (compare $\kappa + \kappa' = 0$ for affine KM and $\kappa + \kappa' = 13$ for Virasoro). $\square$

1.4 Chiral Koszulness: CE versus chiral bar

Proposition 1.4 (PBW Koszulness of the $\mathcal{N} = 2$ SCA;). *The $\mathcal{N} = 2$ superconformal algebra is chirally Koszul at generic central charge. The PBW spectral sequence collapses at E_2 , and $H^n(\overline{B}(\text{SCA}_c)) = 0$ for $n \geq 2$ in the bar-relevant range.*

Proof. The $\mathcal{N} = 2$ SCA is freely strongly generated by $\{T, J, G^+, G^-\}$ at generic c (Adamović [?]: no singular vectors in the vacuum Verma module at generic level). By Proposition ??, free strong generation implies PBW collapse at E_2 , hence chiral Koszulness.

The CE complex of the negative-mode Lie superalgebra $\mathfrak{g}_-$ has $H_{\text{CE}}^2 \neq 0$ at conformal weight 3. This does *not* obstruct chiral Koszulness: the CE complex uses only the mode-0 product $a_{(0)}b$ (the Lie bracket), while the chiral bar differential incorporates all OPE modes via the Orlik–Solomon form factor on configuration spaces (Proposition ??). The sub-leading modes $G_{(1)}^+ G^- = J$ and $G_{(2)}^+ G^- = c/3$ provide additional coboundary contributions in the PBW spectral sequence that kill the CE cocycles at the E_2 page. Verified computationally through conformal weight 6. $\square$

Remark 1.5 (The CE/chiral bar gap). The $\mathcal{N} = 2$ SCA provides the cleanest demonstration in the standard landscape that CE Koszulness (vanishing of $H_{\text{CE}}^2(\mathfrak{g}_-)$) and chiral Koszulness (vanishing of $H^2(\overline{B}(A))$) are genuinely distinct properties. For single-generator algebras (Heisenberg, Virasoro), the two complexes agree at bar degree 2 because the mode-0 product exhausts the generating OPE at that level. For multi-generator algebras with mixed bosonic/fermionic statistics and sub-leading OPE poles, the chiral bar complex is strictly finer. The mechanism: at bar degree 3, the Orlik–Solomon differential on $\text{Conf}_3(\mathbb{C})$ activates the higher OPE modes $G_{(1)}^+ G^-$ and $G_{(2)}^+ G^-$, which are invisible to the CE complex but provide coboundaries in the PBW spectral sequence that annihilate the CE H^2 classes.

1.5 Shadow obstruction tower: multi-channel structure

The $\mathcal{N} = 2$ SCA has four generators spanning three independent primary lines in the cyclic deformation complex $\text{Def}_{\text{cyc}}^{\text{mod}}$ (Definition ??):

- (i) The *T-line* (stress tensor, conformal weight 2, bosonic): identical to the Virasoro line. The *TT* OPE has maximal pole order 4; $d \log$ absorption (Remark ??) gives collision depth $k_{\text{max}} = 3$. Class M: infinite shadow depth.

- (ii) The *J-line* (U(1) current, conformal weight 1, bosonic): identical to the Heisenberg line. The *JJ* OPE has maximal pole order 2, giving $k_{\max} = 1$. Class G: Gaussian, shadow depth $r_{\max} = 2$.
- (iii) The *G-lines* (supercurrents $G^\pm$, conformal weight $\frac{3}{2}$, fermionic): the $G^\pm G^\pm$ OPEs vanish; the mixed $G^+ G^-$ OPE has maximal pole order 3, giving $k_{\max} = 2$. Class L: Lie/tree, shadow depth $r_{\max} = 3$.

The overall shadow class is M (governed by the deepest channel), and the shadow depth is $r_{\max} = \infty$. The modular characteristic $\kappa = (k+4)/4$ receives contributions from all channels via the Kazama–Suzuki coset decomposition (Proposition ??).

Remark 1.6 (Shadow obstruction tower for the $\mathcal{N} = 2$ SCA). For SCA_c at generic level, the shadow obstruction tower in the modular convolution algebra $\mathfrak{g}_A^{\text{mod}}$ (Definition ??) has modular characteristic $\kappa = (k+4)/4$. Since the $\mathcal{N} = 2$ SCA has generators at three distinct conformal weights $(1, \frac{3}{2}, 2)$, it is a multi-weight algebra; the genus-1 identity $\text{obs}_1 = \kappa \cdot \lambda_1$ holds unconditionally, while the higher-genus scalar formula receives cross-channel corrections at $g \geq 2$. The tower does not terminate: the *T*-channel carries the same infinite tower as the Virasoro algebra (Chapter ??), including a nonzero quintic obstruction $o_5^{(T)} \neq 0$. The *J*-channel terminates at arity 2 (Gaussian) and the *G*-channel at arity 3 (Lie/tree, from the $G^+ G^-$ bracket). Cross-channel interactions between the *T*-, *J*-, and *G*-channels produce additional multi-channel shadow coefficients at arities ≥ 4 .

1.6 Spectral flow

The $\mathcal{N} = 2$ SCA admits a one-parameter family of automorphisms σ_θ ($\theta \in \mathbb{Z}$), the *spectral flow*, defined by

$$\sigma_\theta(T) = T + \theta J + \frac{c}{6} \theta^2, \quad \sigma_\theta(J) = J + \frac{c}{3} \theta, \quad \sigma_\theta(G^\pm) = G_{\pm\theta}^\pm, \quad (1.13)$$

where $G_{\pm\theta}^\pm$ denotes the shift of mode indices by $\pm\theta$. At $\theta = 1/2$, the spectral flow intertwines the Neveu–Schwarz and Ramond sectors.

Since σ_θ is an algebra automorphism, it preserves all OPE structure constants and hence every shadow coefficient: $S_r^{\sigma_\theta} = S_r$ for all r . The modular characteristic κ , the cubic shadow C , the quartic resonance class Q , and the full shadow obstruction tower are spectral-flow invariants. This provides a non-trivial consistency check: the NS and Ramond sectors have different field content (integer vs. half-integer moded supercurrents) but identical shadow data (Remark ??).

Remark 1.7 (Comparison with the $\mathcal{N} = 1$ and $\mathcal{N} = 4$ families). The three fundamental superconformal families form a hierarchy under extended super-

symmetry:

$\mathcal{N}$	c	κ	c'	$\kappa + \kappa'$	Self-dual
0 (Vir)	c	$c/2$	$26-c$	13	$c = 13$
1 (SVir)	c	$(3c-2)/4$	$15-c$	$41/4$	$c = 15/2$
2 (SCA)	$3k/(k+2)$	$(k+4)/4$	$6-c$	1	$c = 3$
4 (small)	$6k$	$2k$	$-c-24$	-8	$c = -12$

As $\mathcal{N}$ increases, the complementarity sum $\kappa + \kappa'$ decreases: 13, $41/4$, 1, -8. This reflects the increasing cancellation between bosonic and fermionic contributions to the genus-1 obstruction class: the modular characteristic is an invariant of the full algebra, not of the Virasoro subalgebra alone.